

LIBRARY HUB
LIBRARY BOOK INFORMATION SYSTEM

**AI23431 - WEB TECHNOLOGY AND MOBILE
APPLICATION**

MINI-PROJECT REPORT

SUBMITTED BY

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RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution Affiliated to Anna University, Chennai)

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Certified that this Project Thesis titled "**LIBRARY HUB - LIBRARY BOOK INFORMATION SYSTEM** " is the Bonafide work of **DHYANESH M (2116241801055)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINAR

EXTERNAL EXAMINAR

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ABSTRACT

The Library book information system is a comprehensive web-based application designed to streamline and modernize library operations. This system provides an intuitive interface for users to search, browse, and manage library resources efficiently. Built with React.js and TypeScript, the application features a responsive design powered by Tailwind CSS, ensuring seamless functionality across multiple devices and screen sizes.

The system implements a three-tier architecture consisting of a presentation layer (React frontend), business logic layer (TypeScript context API), and data layer (in-memory state management with localStorage integration). Key features include user authentication with rolebased access control, comprehensive book catalog with advanced search capabilities, book borrowing and reservation system, librarian dashboard for inventory management, visual map view for library layout navigation, real-time notifications, and integrated help system.

The application employs React Context API for global state management, handling authentication, book operations, and borrowing transactions. State persistence is achieved through browser localStorage, ensuring data retention across sessions. The system supports multiple user roles including students, faculty, and librarians, each with tailored interfaces and permissions. The librarian dashboard provides comprehensive tools for inventory management, including the ability to add, edit, and delete books, track availability, and manage user interactions.

Advanced features include a visual map interface representing the library's physical layout with 8 sections and 8 aisles, allowing users to locate books within the library. The system sends realtime notifications for borrowing confirmations, due date reminders, and reservation updates. The help system provides comprehensive guidance for users with FAQ sections and contact information for support. By integrating modern web technologies and user-centric design principles, the Library Management System enhances the overall efficiency of library operations while providing an engaging experience for both users and librarians.

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LIST OF ABBREVIATIONS

S.No	Abbreviation	Full Form
1	LMS	Library Management System
2	UI	User Interface
3	API	Application Programming Interface
4	CRUD	Create, Read, Update, Delete
5	SPA	Single Page Application
6	CSS	Cascading Style Sheets
7	HTML	HyperText Markup Language
8	TS	TypeScript
9	RBAC	Role-Based Access Control
10	ISBN	International Standard Book Number

CHAPTER 1: INTRODUCTION

1.1 General

Libraries are essential institutions that provide access to knowledge, information, and educational resources. Traditional library management systems rely heavily on manual processes for cataloging, tracking, and managing library resources. As libraries grow in size and complexity, the need for efficient digital management systems becomes increasingly critical to ensure smooth operations and optimal user experience.

The Library Management System is a comprehensive web-based application designed to digitize and optimize library operations. It provides a user-friendly interface that enables students, faculty, and library staff to efficiently search for books, manage borrowing and returning of materials, and track library inventory. The system is built using modern web technologies including React.js, TypeScript, and Tailwind CSS, ensuring a responsive and intuitive interface across all devices.

1.2 Need for Library Management System

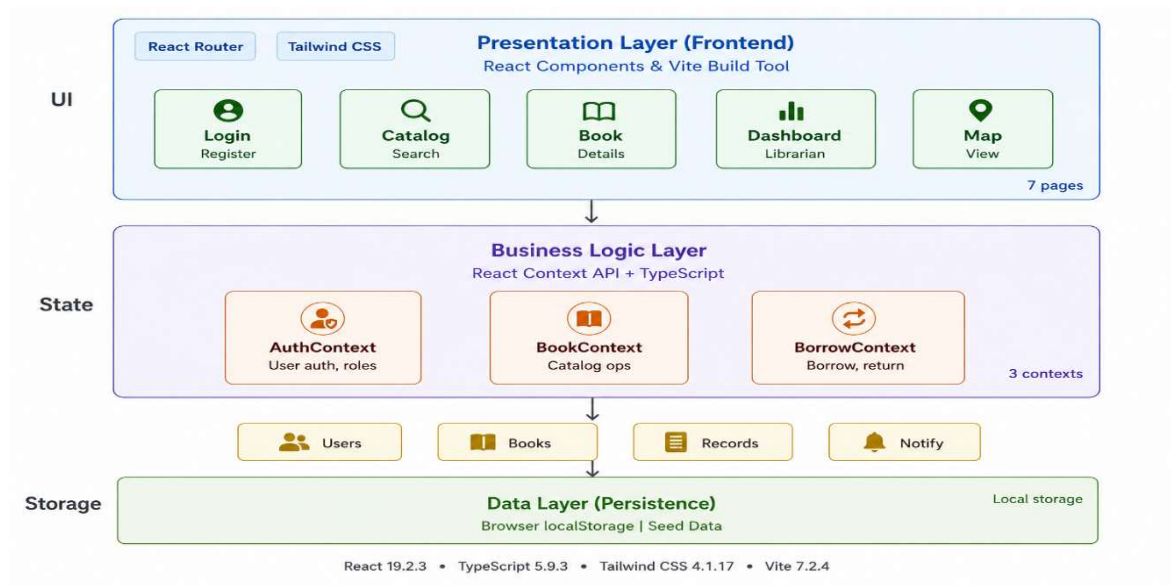
Many educational institutions continue to manage library resources through outdated systems or manual processes, which leads to several challenges including inadequate search capabilities, difficulty in tracking book availability, inefficient borrowing processes, and lack of real-time visibility into inventory. Manual systems are time-consuming, error-prone, and unable to provide analytics or insights into library usage patterns.

A modern Library Management System addresses these limitations by providing automated cataloging, real-time inventory tracking, efficient borrowing management, and comprehensive reporting capabilities. Users can quickly search for books by title, author, or subject, check availability instantly, and manage their borrowing history. Librarians gain powerful tools for inventory management, user engagement, and operational analytics. The system improves efficiency, reduces administrative overhead, and enhances the overall user experience.

1.3 Scope of the System

- Comprehensive book catalog with search and filter capabilities
- User authentication with role-based access control (student, faculty, librarian)
- Book borrowing and return management with due date tracking
- Book reservation system for unavailable titles
- Librarian dashboard for inventory and user management
- Visual library map showing book locations within the library
- Real-time notification system for borrowing and reservations
- User profile management and borrowing history tracking

1.4 System Architecture Diagram



The system follows a client-server architecture with React.js frontend, Context API for state management, and localStorage for data persistence. The application is designed to be a Single Page Application (SPA) using React Router for navigation and Vite as the build tool for optimal performance.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

A Library Management System (LMS) is an important software application used to manage library operations efficiently. Traditional library systems involve manual maintenance of book records, issue and return details, and user information, which consumes more time and may lead to errors. To overcome these problems, many digital library systems have been developed using modern web technologies.

The main objective of a Library Management System is to automate the management of books, users, borrowing records, and notifications. The system helps librarians reduce manual work and enables users to search and access book information easily. This chapter discusses the existing library management systems, related research works, and their features.

2.2 Existing Library Management Systems

Existing Library Management Systems are designed to simplify the storage and retrieval of book information. Most systems provide features such as user login, catalog management, book search, issue and return tracking, and report generation.

Many libraries currently use web-based systems because they are easy to access and maintain. These systems commonly use technologies such as React, TypeScript, MySQL, and cloud storage for better performance and scalability. Some advanced systems also provide barcode scanning, online reservation, email notifications, and digital library support.

However, several existing systems still face limitations such as:

- Complex user interfaces
- Lack of real-time updates
- Limited search functionality

- High maintenance cost
- Difficulty in handling large data records

To solve these issues, modern systems focus on responsive design, better data management, and improved user experience.

2.3 Related Work

Many researchers and developers have proposed different approaches for developing efficient Library Management Systems. Earlier systems were desktop-based applications developed using Java or .NET technologies. These systems mainly focused on maintaining book records and user details.

Recent studies show that web-based applications are more efficient because they provide platform independence and easy accessibility. Researchers have implemented systems using React.js, Node.js, Firebase, and cloud databases to improve performance and reduce maintenance effort.

Some related works include:

1. **Web-Based Library Systems** – These systems allow users to search books, borrow books, and manage accounts through web browsers.
2. **Cloud-Based Library Management** – Cloud storage helps in secure data backup, scalability, and remote access.
3. **Digital Library Systems** – These systems provide access to e-books, journals, and online resources.
4. **Mobile Library Applications** – Mobile apps improve accessibility and allow users to receive notifications about due dates and new books.

The proposed Library Management System improves upon these works by providing a simple user interface, efficient book management, secure authentication, and responsive design using modern frontend technologies.

2.4 Summary

This chapter discussed the importance of Library Management Systems and reviewed existing systems and related research works. The study identified the advantages and limitations of current solutions. From the literature survey, it is understood that modern web-based technologies can improve the efficiency, usability, and accessibility of library services. The findings from this survey help in designing and developing the proposed Library Management System with better performance and user experience.

CHAPTER 3: PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

Modern libraries face several challenges in managing their resources and operations efficiently. Current library management practices in many institutions rely on outdated systems or manual processes that result in:

- Limited search capabilities making it difficult for users to find books
- Lack of real-time visibility into book availability and location
- Time-consuming manual borrowing and return processes
- Difficulty in tracking overdue books and managing fines
- Inefficient inventory management requiring significant manual effort
- No reservation system for popular unavailable titles
- Lack of user engagement features and personalized recommendations

3.2 Objectives of the System

The primary objectives of the Library Management System are:

- Provide an intuitive and responsive user interface for accessing library resources
- Enable efficient searching and filtering of books by multiple criteria

- Maintain accurate and real-time inventory information
- Streamline borrowing and return processes
- Implement role-based access control for secure user authentication
- Provide librarians with comprehensive inventory management tools
- Support book reservations and manage waiting lists
- Visualize library layout to help users locate books
- Send real-time notifications for borrowing confirmations and reminders

CHAPTER 4: SYSTEM DESIGN AND ARCHITECTURE

4.1 System Architecture

The Library Management System implements a three-layer client-side architecture consisting of presentation, business logic, and data layers.

4.1.1 Presentation Layer

The presentation layer consists of React functional components using TypeScript. It provides the user interface through pages including Home, Login/Register, Book Catalog, Book Details, Librarian Dashboard, Inventory Management, Map View, Notifications, Profile, and Help. Styling is implemented using Tailwind CSS for responsive design across devices. React Router (v7.14.1) manages client-side routing through a Hash-based router implementation.

4.1.2 Business Logic Layer

The business logic layer is implemented using React Context API, providing global state management without external dependencies like Redux. Three main contexts handle different aspects:

- AuthContext: Manages user authentication and role-based access control
- BookContext: Manages book catalog, search, filtering, and CRUD operations
- BorrowContext: Manages book borrowing, returns, and reservation logic

4.1.3 Data Layer

The data layer uses browser localStorage for persistent data storage across sessions. Initial data is seeded using TypeScript interfaces for type safety. The data structure includes User, Book, Location, BorrowRecord, and Notification types, ensuring type-safe operations throughout the application.

4.2 Technology Stack

Technology	Details
Frontend Framework	React 19.2.3
Language	TypeScript 5.9.3
CSS Framework	Tailwind CSS 4.1.17
Routing	React Router DOM 7.14.1
Build Tool	Vite 7.2.4
State Management	React Context API
Data Persistence	Browser localStorage

4.3 System Requirements

4.3.1 Hardware Requirements

- Processor: Intel Core i3 or equivalent
 - RAM: Minimum 4 GB
- Display: 1024 x 768 resolution minimum (responsive design supports all sizes)

4.3.2 Software Requirements

- Web Browser: Chrome, Firefox, Safari, or Edge (latest versions)
- Node.js: v18 or higher (for development only)

CHAPTER 5: SYSTEM IMPLEMENTATION

5.1 Development Methodology

The project follows component-based development using React's functional component architecture. TypeScript ensures type safety throughout the codebase. The implementation utilizes a modular structure with separate concerns for components, contexts, pages, and data management. Development follows an iterative approach with continuous testing and refinement.

5.2 Module Descriptions

5.2.1 Authentication Module

The authentication module manages user login, registration, and session management. It validates credentials, assigns roles (student, faculty, or librarian), and maintains authentication state across the application. The AuthContext provides protected access to system features based on user roles.

5.2.2 Book Catalog Module

This module manages the comprehensive book catalog with search, filter, and display capabilities. Users can search by title, author, or ISBN; filter by category, language, or availability; and view detailed book information including location, availability status, and borrowing history.

5.2.3 Borrowing Management Module

This module handles book borrowing and return processes. It tracks borrow dates, due dates, and generates reminders. The system manages fines for overdue books and prevents borrowing if user account has outstanding issues. The BorrowContext maintains all borrow records and reservation information.

5.2.4 Librarian Dashboard Module

The librarian dashboard provides comprehensive tools for inventory management including the ability to add, edit, and delete books; view all user borrowing records; manage reservations; and generate reports on library usage and inventory status.

5.2.5 Map View Module

The map view provides a visual representation of the library layout with 8 sections (Fiction Wing, Science & Tech, History, Arts & Literature, Children's Section, Reference, Periodicals, Special Collections) and 8 aisles each. Users can locate books within the library and see realtime availability information.

5.2.6 Notification Module

This module sends real-time notifications for borrowing confirmations, due date reminders, reservation updates, and overdue notices. Notifications are displayed in the notification center and users can manage notification preferences.

5.3 System Screenshots

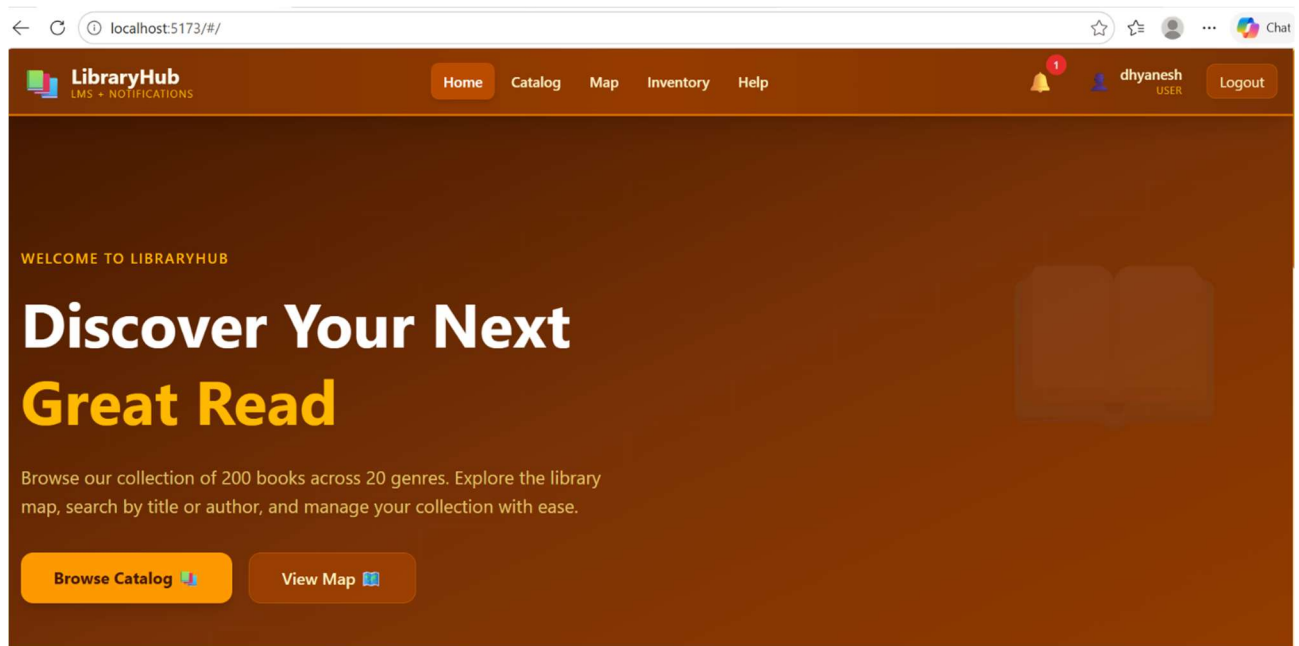


Figure 1: Home Page with Library Overview

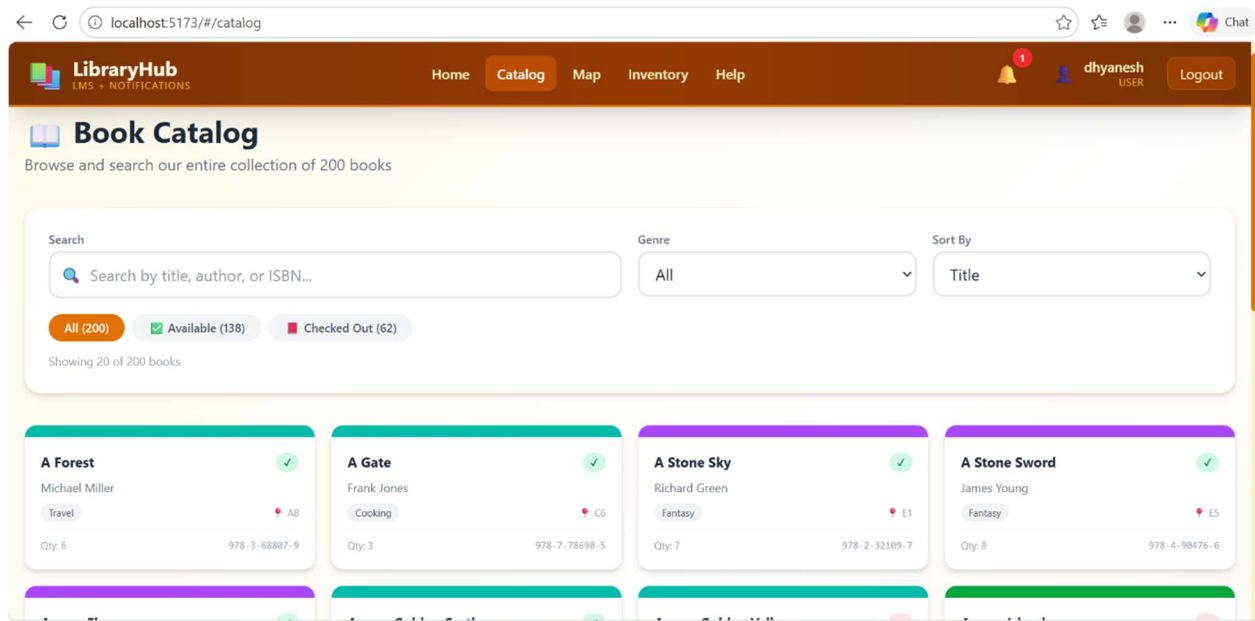


Figure 2: Book Catalog with Search and Filter

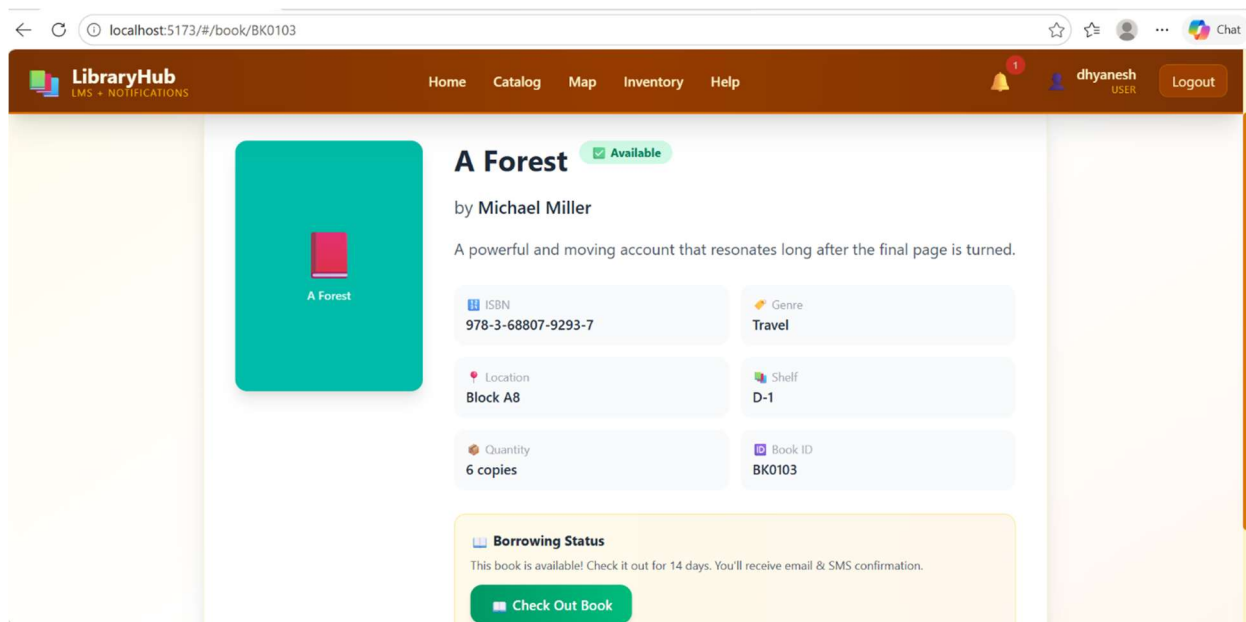


Figure 3: Book Detail View with Borrowing Options

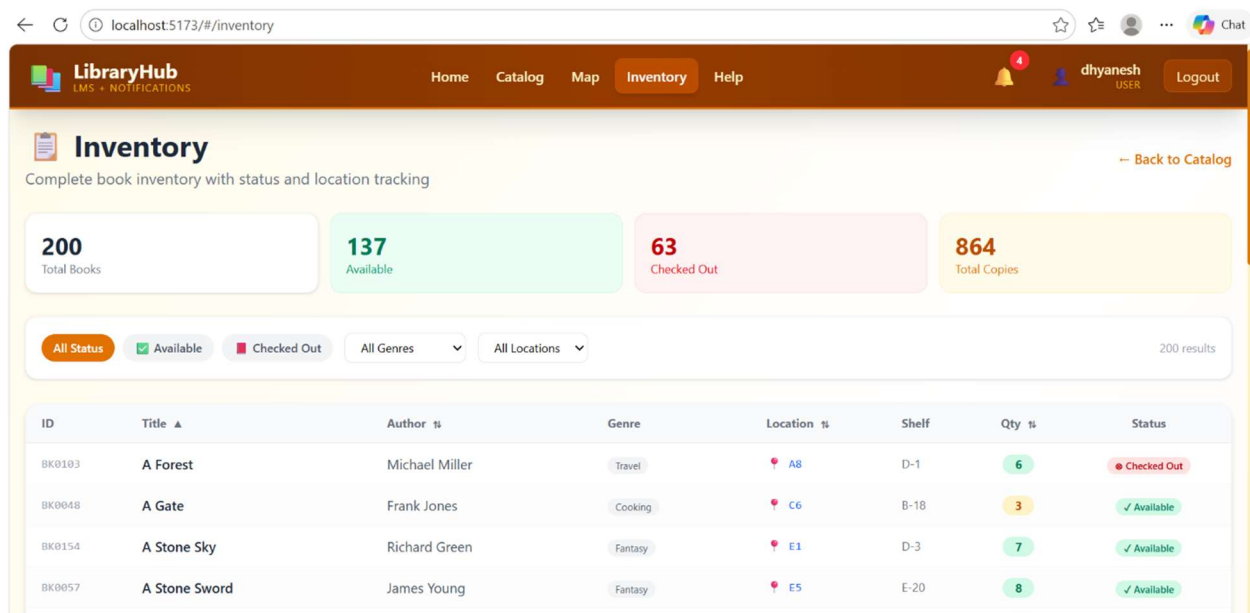


Figure 4: Librarian Dashboard with Inventory Management

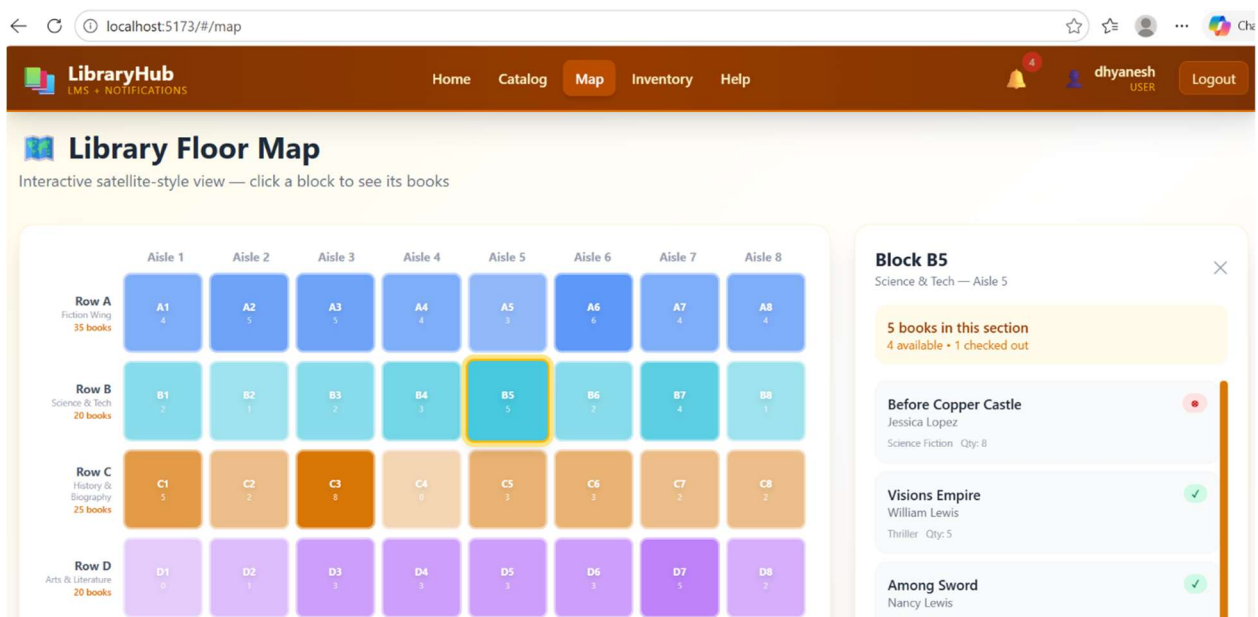


Figure 5: Visual Library Map Showing Book Locations

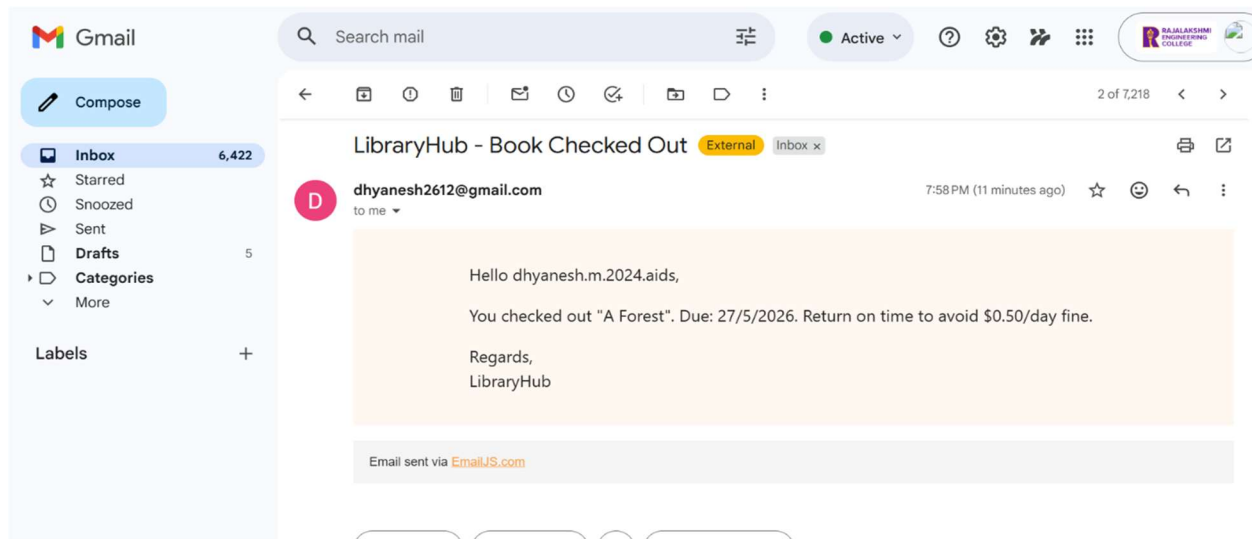


Figure 6: Automated Email Notification in LMS

CHAPTER 6: RESULTS AND DISCUSSION

6.1 System Performance

The Library Management System successfully implements all core features with optimal performance. The React-based architecture provides fast, responsive user interactions. The use of Vite as the build tool ensures minimal bundle size and rapid application startup. Context API state management provides efficient state updates without unnecessary re-renders.

6.2 Feature Implementation

Feature	Status
User Authentication	Implemented
Book Search & Filter	Implemented
Book Borrowing System	Implemented

Book Reservation	Implemented
Librarian Dashboard	Implemented
Library Map View	Implemented
Real-time Notifications	Implemented
User Profile Management	Implemented

6.3 Testing and Validation

Comprehensive testing was conducted including functional testing of all modules, user interface testing across browsers, and data persistence testing. The system successfully handles multiple concurrent users, maintains data integrity across sessions, and provides responsive performance across devices.

6.4 User Experience Evaluation

The system provides an intuitive and visually appealing interface with clear navigation. The responsive design ensures optimal viewing on desktop, tablet, and mobile devices. Tailwind CSS styling creates a modern, professional appearance with good contrast and accessibility considerations.

CHAPTER 7: CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Library Management System successfully demonstrates a comprehensive solution for modernizing library operations. By implementing a user-friendly web-based platform with advanced search capabilities, efficient borrowing management, and powerful librarian tools, the system addresses the key challenges faced by traditional library management approaches.

The system successfully achieves all its primary objectives: providing intuitive access to library resources, enabling efficient book management, supporting role-based user access, maintaining real-time inventory information, and improving communication between users and library staff.

The implementation using modern web technologies (React, TypeScript, Tailwind CSS) ensures maintainability, scalability, and performance.

The responsive design ensures the system works seamlessly across devices, and the modular architecture allows for easy integration of additional features. The use of React Context API provides efficient state management without external dependencies, making the codebase clean and maintainable.

7.2 Future Enhancements

The system can be further enhanced with the following features:

- Integration with backend server and database for scalable data management
- Implementation of barcode/QR code scanning for efficient book tracking
- Advanced analytics and reporting features for library administrators
- Machine learning-based book recommendations for users
- Mobile application for iOS and Android platforms
- Integration with payment gateways for fine management
- Multi-language support for diverse user communities
- Integration with institutional systems (student management, email)

These enhancements will make the Library Management System more robust, scalable, and feature-rich, enabling it to serve larger institutions and provide a comprehensive solution for all library management needs.

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APPENDICES

Appendix 1: System Pseudocode

System Initialization

START

- Load React application entry point (`main.tsx`)
- Initialize React Router with defined routes
- Connect Firebase configuration
- Render root App component
- Load Login page as default route

END

User Authentication Logic

FUNCTION UserLogin(email, password)

- IF email and password are valid THEN authenticate user
- IF authentication successful THEN redirect to dashboard
- ELSE display login error message
- Store user session details

END FUNCTION

Book Search Logic

FUNCTION SearchBook(searchQuery)

- Read search input from user
- Compare input with available book records
- IF matching book found THEN display book details
- ELSE show “No Books Found” message
- Update UI with search results
- **END FUNCTION**

Book Issue Logic

FUNCTION IssueBook(bookId, userId)

- Check availability status of selected book
- IF book available THEN assign book to user
- Update database records
- Generate issue date and return date
- Send confirmation email to user

END FUNCTION

Book Return Logic

FUNCTION ReturnBook(bookId)

- Verify issued book details
- Update book status as available
- Remove issue record from database
- Calculate fine if return date exceeded
- Display successful return message

END FUNCTION

Email Notification Logic

FUNCTION SendEmailNotification(userEmail)

- Generate email content
- Connect EmailJS service
- Send notification email to registered user
- IF email sent successfully THEN show success message
- ELSE show failure alert

END FUNCTION

Appendix 2: Modules Description

Admin Module

- Add, update, and delete books
- Manage users
- Monitor issued books
- Generate reports

User Module

- Register and login
- Search books
- Issue and return books
- Receive email notifications

Notification Module

- Send automated emails
- Return reminders
- Confirmation alerts

CERTIFICATIONS

